Chapter 1

INTRODUCTION

A **crankshaft** is a critical component in internal combustion engines, converting the linear motion of the pistons into rotational motion that drives the vehicle's wheels or other machinery. It consists of a long, rotating shaft with several bends and offsets, designed to withstand significant mechanical stresses, including compression, tension, and torsion.

A **special purpose gauge** for crankshafts is a highly specialized measuring instrument used in the manufacturing and maintenance processes to ensure the accuracy and precision of crankshaft dimensions. These gauges are custom-designed to measure specific features of the crankshaft, such as:

A **special purpose gauge** for crankshafts is a precision instrument designed specifically to measure and ensure the accuracy of crankshaft dimensions during manufacturing or maintenance processes. Crankshafts, being integral components of internal combustion engines, need to meet very stringent tolerances in order to function efficiently and reliably. The special purpose gauge plays a vital role in achieving this by providing highly accurate measurements of critical features like the diameter of journals, crankpins, and overall straightness.

Due to the complex shape of a crankshaft, traditional measuring tools may not be suitable for precise and quick measurements of all its features. Special purpose gauges are engineered to measure these intricate aspects efficiently while ensuring quality control and minimizing errors in mass production or during maintenance.

Who invented the special purpose gauge

The invention of the **special purpose gauge** cannot be attributed to a single individual, as it is the result of continuous advancements in precision measurement technology. However, the development of gauges in general, including special purpose gauges, is closely linked to the evolution of precision engineering and the industrial revolution.

Key Historical Figures and Developments:

1. **Joseph Whitworth (1803-1887)**: Joseph Whitworth, an English engineer, made significant contributions to precision engineering and standardized measurements. He is often credited with the development of the first widely used precision measurement standards in the mid-19th century. Although he didn't specifically invent special purpose gauges, his work in establishing consistent measurement standards laid the foundation for the development of specialized gauges.

2. **Henry Maudslay (1771-1831)**: Another influential figure in the development of precision tools, Maudslay made important advances in the manufacturing of machine tools. His work on screw threads and measuring instruments contributed to the creation of more accurate measurement tools, which would later include specialized gauges.

3. **The Rise of Precision Engineering (Late 19th and Early 20th Century)**: The widespread adoption of **interchangeable parts** during the Industrial Revolution, and later advancements in automotive, aerospace, and machinery manufacturing, necessitated the development of specialized measuring instruments. This period saw the rise of **micrometres**, **callipers**, and **gauges**, which were used to measure various components with precision.

4. **The Development of Go/No-Go Gauges and Custom Measuring Tools**: As manufacturing processes became more complex, especially with the rise of automotive and engine manufacturing, engineers began creating **special purpose gauges** tailored to specific components like crankshafts, camshafts, and other machine parts. These gauges were often developed by companies and engineers involved in specific industries, making it difficult to attribute the invention of these tools to a single individual.

1.2 Equipment

Special purpose gauges are designed for specific measurement tasks, typically for assessing the dimensions and features of components with high precision. The equipment used in the creation, calibration, and operation of these gauges includes various tools and technologies to ensure

accuracy, efficiency, and reliability. Here's an overview of the key equipment used for special purpose gauges:

1.3 Measuring Instruments

A. Micrometres:

- Micrometres are precision tools used to measure the dimensions of a crankshaft's journals, pins, or other critical features.
- These devices are highly accurate and can measure down to thousandths of a millimetre or even microns.

B. Callipers:

- Digital or analogue callipers are used to measure the internal and external dimensions of crankshafts, pins, and other components.
- They provide quick and accurate readings, suitable for various measuring tasks.

C. Dial Indicators:

- Used primarily to measure **run-out**, which is the deviation of the crankshaft from the true centreline or axis of rotation.
- Dial indicators are also used to measure straightness and alignment in components like crankshafts.

D. Depth Gauges:

- These are used to measure the depth of keyways, holes, or slots in crankshafts or other machined parts.
- Depth gauges ensure precision in features that require depth consistency.

E. Coordinate Measuring Machines (CMMs):

- CMMs are advanced equipment that use a probe (mechanical, laser, or optical) to measure the 3D geometry of a part.
- They provide high-accuracy measurements and are used for complex parts, including crankshafts, to verify their dimensions in all three axes.

1.4 Special Purpose Gauges

A. Go/No-Go Gauges:

- These are simple gauges that check whether a part meets specified tolerances.
- A **Go gauge** confirms that the part fits within the acceptable limits, while a **No-Go gauge** ensures that the part does not exceed maximum tolerances.
- These are commonly used for rapid inspections during mass production.

B. Plug and Ring Gauges:

- **Plug gauges** are used to measure the internal diameter of holes or bores, such as the holes in crankshaft journals.
- **Ring gauges** are used for external diameter measurements, ensuring the crankshaft pins and journals are within specified limits.

C. Thread Gauges:

- These gauges are used to inspect threads on parts of the crankshaft, ensuring that threaded sections such as bolt holes meet the required specifications.

D. Profile Gauges:

- Used to measure the contour or profile of a part. These gauges are essential when measuring the complex profiles of crankshafts or camshafts.

1.5 Inspection Equipment

A. Surface Plate:

- A flat, stable platform used for inspecting the flatness and straightness of crankshafts and other components.
- It serves as a reference for precise measurements and ensures that the components are aligned for measurement.

B. Optical Comparators:

- Optical comparators use light to project the shadow of a part onto a screen, allowing operators to compare its profile with a reference or blueprint.
- These are useful for checking the shape and form of complex parts like crankshafts.

C. Laser Scanning Systems:

- Laser scanners are used to capture the 3D geometry of a part, such as a crankshaft, to assess its dimensions and surface quality.
- Laser scanning provides high-precision data that can be used for detailed analysis.

D. X-Ray or Ultrasound Inspection:

- These methods can be used to inspect the internal quality of crankshafts for flaws like cracks or voids.
- Non-destructive testing ensures that the material integrity of the crankshaft is intact.

1.6 Calibration Equipment

A. Calibration Blocks:

- These are precision reference blocks used to calibrate measuring instruments like micrometres, callipers, and CMMs.
- Calibration ensures that gauges provide accurate measurements throughout their usage.

B. Gauge Blocks:

- Gauge blocks are used to set the standard for length measurements and calibrate various tools, including special purpose gauges.
- They help maintain the accuracy of precision measuring equipment.

C. Vernier Calibration Tools:

- These tools are used for verifying the accuracy of other measurement instruments, ensuring they remain precise and within tolerance.

5. Computer-Aided Inspection (CAI) Tools

A. CAD Software for Inspection:

- CAD software is used to design and inspect the geometry of parts like crankshafts digitally.
- It can simulate the measuring process, ensuring that parts meet all design specifications.

B. Data Acquisition Systems:

- These systems collect and analyse data from measuring instruments, allowing for real-time monitoring of the inspection process.
- They are used to track measurements, generate reports, and compare data with the required specifications.

Process Parameters and Machining Characteristics of Special Purpose Gauges

1. Machining Characteristics of Special Purpose Gauges

The machining characteristics refer to the properties of the materials being machined and the effects of various machining processes on them. For special purpose gauges, the machining characteristics must be controlled to ensure high precision, reliability, and functionality.

A. Material Selection:

- Special purpose gauges are typically made from **high-quality tool steels**, **hardened materials**, or **carbides** to ensure durability and resistance to wear.
- Materials used in gauge production must have stable dimensional characteristics and resistance to thermal expansion, ensuring that the gauge remains accurate over time and under various environmental conditions.

B. Surface Finish:

- Achieving a **smooth surface finish** is critical in the manufacturing of special purpose gauges. The surface finish affects the gauge's ability to measure with precision and ensures minimal friction and wear.
- Special purpose gauges often require a **mirror-like finish**, especially on contact points, to guarantee accurate measurement and prolonged service life.

C. Tolerance and Accuracy:

- **Tolerances** for special purpose gauges are typically very tight, sometimes to the micrometres or even manometer level, especially for measuring high-precision components like crankshafts.
- **Machining precision** must meet strict limits to ensure that the gauges provide accurate measurements. This includes minimizing errors caused by thermal expansion, vibration, and tool wear.

D. Measurement Stability:

- The **stability of the gauge** during measurements is critical. This includes not only the mechanical stability of the gauge but also its **environmental stability** (e.g., temperature stability).
- Special purpose gauges are often designed to be **thermally stable** and **insensitive to environmental changes** to ensure consistent readings in different working conditions.

E. Dimensional Accuracy:

- Special purpose gauges must be produced with **extremely tight dimensional tolerances** to meet the precision standards required for measuring complex components.
- **Micron-level accuracy** is often necessary, especially when the gauge is used for measuring parts such as crankshafts, which require fine precision for successful functioning in engines.

3. Common Machining Processes for Special Purpose Gauges

Special purpose gauges are manufactured using a variety of advanced machining processes, each

designed to achieve the necessary precision and surface quality:

A. CNC Turning and Milling:

- These processes are often used for creating complex shapes and features, such as the cylindrical profiles of crankshaft gauges.
- Computer numerical control (CNC) machines allow for extremely precise control over cutting speed, feed rate, and depth of cut.

B. Grinding:

- **Surface grinding** and **cylindrical grinding** are used to achieve fine finishes and precise dimensions on gauge components.
- Grinding is essential for achieving the mirror-like finish and high accuracy required in special purpose gauges.

➢ ADVANTAGES

1. High Precision and Accuracy

- **Tailored Measurement**: Special purpose gauges are specifically designed to measure complex crankshaft features such as journal diameters, crankpin diameters, taper, ovality, and run-out. This level of customization ensures that measurements are highly accurate, often to micron-level precision.
- **Consistency**: These gauges provide consistent and repeatable measurements, reducing human error and ensuring that all crankshafts meet the same exacting standards.

2. Improved Quality Control

- **Instant Feedback**: Special purpose gauges, such as offer fast, reliable assessments of crankshaft dimensions, ensuring that parts that fall outside acceptable tolerances are immediately identified and rejected.
- **Minimized Defects**: By accurately measuring critical features, these gauges help identify defects such as misalignment, irregular wear, or dimensional inaccuracies that could lead to engine failure or reduced performance.

3. Increased Efficiency in Production

- **Faster Inspection**: Compared to traditional measurement methods, special purpose gauges allow for quicker inspection of crankshafts. This leads to faster turnaround times in mass production environments, improving overall productivity.
- **Reduced Downtime**: Because these gauges are designed for specific tasks, they reduce the time spent on manual inspections or adjusting machinery. This results in more efficient manufacturing processes.

4. Cost-Effectiveness

- **Reduced Waste**: By ensuring that only crankshafts within specification are passed on to the next stages of production, special purpose gauges reduce the number of defective parts, leading to less waste and material costs.
- **Lower Rework Costs**: With highly accurate measurements, the likelihood of having to rework crankshafts due to dimensional errors is significantly reduced, saving both time and resources.

5. Customization for Specific Applications

- **Adaptability**: Special purpose gauges can be tailored to measure specific features or geometries of different crankshaft designs. Whether for automotive, industrial, or high-performance engines, these gauges can be customized for each type of crankshaft, ensuring optimal performance and compatibility.
- **Flexibility**: The gauges can be designed to address particular measurement challenges, such as measuring parts with complex or intricate geometries that standard measuring tools cannot assess effectively.

6. Reduction of Human Error

- **Automated and Simplified Measurement**: Many special purpose gauges, such as are simple to use and require little operator intervention. This reduces the chance for human error, especially in high-volume production, where fatigue or oversight can lead to incorrect readings.
- **User-Friendly Operation**: These gauges are typically designed for ease of use, with clear indicators and simple measurement procedur.

7. Non-Destructive Testing

- **Avoids Damage to Parts**: Special purpose gauges are non-destructive, meaning they don't damage the crankshaft during inspection. This is especially important when measuring high-value or sensitive components, as it preserves the integrity of the parts being measured.

❖ COMPANY PROFILE

Table No:1.1: Company Profile

Company Name	**VINAYAK FOUNDRY**
Year of Establishment	2011
Legal Status of Firm	Partnership Firm
Owners	Mr. Vinayak Patil Mr. Narayan Patil
Nature of Business	Manufacturer
Number of Employees	50
Annual Turnover	5-7 crore
India MART member since	June 2012
Registered Address	Plot no.390/3B , Peeranwadi Belgaum Peeranwadi Belgaum-590014 Karnataka
Website link	https://www.indiamart.com/vinayak-founders-aboutus.html

HISTORICAL BACKGROUND OF THE COMPANY

Vinayak Foundry in Machhe, Belgaum is known to satisfactorily cater to demands of its customer base. The business came into existence in 2011 and has since then been a known name in its field. It stands located at wag wade Road, Machhe, Machhe-590014. The business strives to make for a positive experience through its offerings. The accepted modes of payment such as cheque/DD (Demand Draft), cash, cash make every business transaction easy and seamless contributing to making the entire process even more effective.

Assemble components for the both small and big projects, such as machine. To do their work precisely, they use a wide variety of instruments, including blowtorches, shears, gauges, nail sets, and presses. To work in the field, a person needs to complete a degree programme.

Vinayak Foundry in Machhe, Belgaum was established in 2011.

Customer centricity is at the care of a Vinayak Foundry in Machhe, Belgaum and it is this belief that has led the business to build long term relationships. Ensuring a positive customer experience, making available goods and/or services that are of top-notch quality is given prime importance.

India's leading B2B market place, Jd Mart ensures engaging in business activities is a seamless process for small and medium enterprises as well as large businesses in a wake to enable these businesses to reach their audience, this portal lets them showcase their offerings in terms of the products and/or services through a digital catalogue.

Chapter 2

OBJECTIVE(S) OF THE PROJECT

Company Name: VINAYAK FOUNDRY PVT LTD

2.1 Problem Statement:

The processes are taking more time to supervise the process done on each side of the job so here we are trying to minimize the time required to check the operation on the job or product.

2.2 OBJECTIVES

- To reduce the cycle time of the inspection.
- To increase the production.
- To measure and control the field parameters to increase safety efficiency of the process.

Chapter 3

LITERATURE REVIEW

Ozlem Semar This study is an overview of the capability evaluation of measurement systems. Principally, the determination of capability of a measurement system is an important aspect of quality and process improvement initiatives. In practice, various methods are developed and used for determining measurement system capability. As a measurement tool, potential effectiveness of gauge should be considered and significant factors should be identified for that purpose. A set of procedures referred to as Measurement Systems Capability Studies are conducted for assessing capability of gauge, isolating sources of variability in the system, evaluating how much of total observed variability is due to gauge, and investigating two components of measurement error: repeatability and reproducibility of gauge. Gauge Repeatability & Reproducibility (Gauge R&R) Study tries to estimate repeatability and reproducibility components of measurement system variation with primary objective of assessing whether gauge is suitable for intended application or not. Measurement System Analysis (MSA) is a collection of statistical methods, which includes Gauge R&R Study, for analysis of measurement system capability. In this study, detailed literature review of MSA, Gauge R&R and Measurement Systems Capability Studies, and general discussion of misclassification probabilities that give useful, reliable information about measurement systems performance would be provided.

Joel Armon Measurement system analysis is crucial in the continuous improvement of manufacturing processes. Gauge Repeatability and Reproducibility (GR&R) is a process of examining the conformity or non-conformity of the major components of a measurement system; tools, equipment, and operators. This reveals the component contributing the most error or variation in the measurement system. This paper reviews GR&R in dimensional measurements with particular emphasis on accuracy, precision and reliability. There is an introduction to metrology and some metrology terminologies to lay the foundation for the appreciation of gauge repeatability and reproducibility. A critical review of some literature on GR&R studies is done and is expanded to cover general dimensional measurement and investigation of two main areas of measurement error in gauges. The effectiveness of a gauge as a measuring tool is crucial in determining the gauge capability thus, this GR&R review

S. Ochoa-Rodriguez1,2, L.-P. Wang1,3, P. Willems4,5, and C. Onof Radar-rain gauge merging techniques have been widely used to improve the applicability of radar and rain gauge rainfall estimates by combining their advantages, while partially overcoming their individual weaknesses.

Despite significant research in this area, guidance on the suitability of and factors affecting merging techniques at the fine spatial-temporal resolutions required for urban hydrological applications is still insufficient. In this paper, an in-depth review of radar-rain gauge merging techniques is conducted, with a focus on their potential for urban hydrological applications. An overview is first given of existing merging techniques and an application-oriented categorization is proposed: (1) radar bias adjustment methods, (2) rain gauge interpolation methods using radar spatial association as additional information, and (3) radar-rain gauge integration methods. A detailed review is given of studies focusing on the evaluation and intercomparison of merging methods, based upon which the most widely used and best performing techniques from each category are identified. These are mean field bias adjustment, kriging with external drift, and Bayesian merging. Climatological, operational, and methodological factors affecting merging performance are then reviewed and their relevance for urban applications discussed. Based on this review, conclusions on merging potential for urban applications are drawn and research gaps are

Identified, which should be addressed to provide further guidance on the use of merging techniques for urban hydrological applications.

Chien-Wei Wu a, W.L. Pearn b, Samuel Kotz c Process capability indices (PCIs), Cp, Ca, and Cepek have been developed in certain manufacturing industry as capability measures based on various criteria, including process consistency, process departure from a target, process yield, and process loss. It is noted in certain recent quality assurance and capability analysis works that the three indices, provide the same lower bounds on the process yield. In this paper, we investigate the behavior of the actual process yield, in terms of the number of non-conformities (in ppm), for processes with fixed index values of possessing different degrees of process centering. We also extend Johnson's [1992. The relationship of to squared error loss. Journal of Quality Technology 24, 211–215] result formulating the relationship between the expected relative squared loss and PCIs. Also, a comparison analysis among PCIs is carried out based on various criteria. The result illustrates some advantages of using the index over the indices and in measuring process capability (yield and loss), since always provides a better protection for the customers. Additionally, several extensions and applications to real world problem are also discussed. The paper contains some material presented in the Kotz and Johnson [2002. Process capability indices—a review, 1992–2000. Journal of Quality Technology 34(1), 1–19] survey but from a different perspective. It also discusses the more recent developments during the years 2002–2006.

Evangelos Cholangitis, MD, Marco Sandolo, MD, Richard Standish, MD, Laura Marelli, MD, Alberto Quaglia, MD, David Patch, MD, Amar P. Dhillon, MD, Andrew Kenneth Burroughs Characteristics for an optimal liver biopsy specimen were recently defined as 20 to 25 mm long and/or containing more than 11 complete portal tracts (CPTs). A systematic review of percutaneous liver biopsy (PLB) and trans jugular liver biopsy (TJLB) series yielded only 32 PLB studies in which these characteristics were evaluated: mean ± SD length, 17.7 ± 5.8 mm and number of CPTs, 7.5 ± 3.4; and 15 TJLB studies: mean ± SD length, 13.5 ± 4.5 mm and number of CPTs, 6.8 ± 2.3. Studies of sampling heterogeneity and interobserver and interobserver variability also used inadequate specimens by present standards. Only 11 (5.3%) of 207 therapeutic studies for chronic hepatitis B and C documented length and/or number of Cs. Of the current 12 studies evaluating non-invasive fibrosis tests, only 8 documented length or number of CPTs, and only 1 documented length and number of Cs. New studies are needed based on adequate liver biopsy samples to provide reliable estimation of grading and staging in chronic liver disease.

Arevalo-Rodriguez I, Muñoz L, Post-dural puncture headache (PDPH) is one of the most common complications of diagnostic and therapeutic lumbar punctures. PDPH is defined as any headache occurring avert a lumbar puncture that worsens within 15 minutes of sitting or standing and is relieved within 15 minutes of the patient lying down. Researchers have suggested many types of interventions to help prevent PDPH. It has been suggested that aspects such as needle tip and gauge can be modified to decrease the incidence of PDPH. Objectives to assess the elects of

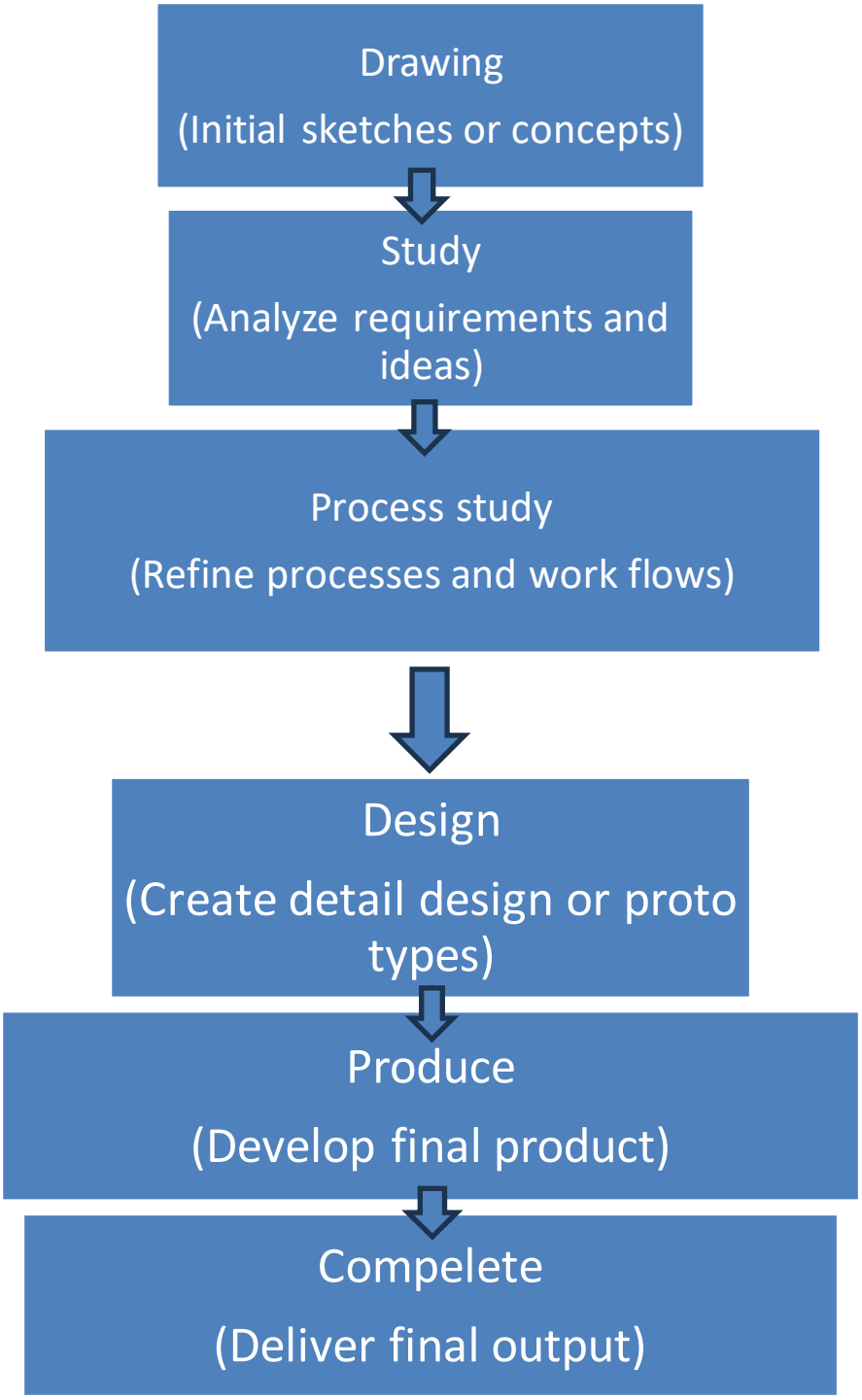
Needle tip design (traumatic versus atraumatic) and diameter (gauge) on the prevention of PDPH In participants who have undergone dural puncture for diagnostic or therapeutic causes. Search methods we searched CENTRAL, MEDLINE, Embase, CINAHL and LILACS, as well as trial registries via the World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP) search portal in September 2016. We adopted the MEDLINE strategy for searching the other databases. The search terms we used were a combination of thesaurus-based and free-text terms for both interventions (lumbar puncture in neurological, anaesthesia or

Pyelography settings) and headache. Selection criteria we included randomized controlled trials (RCTs) conducted in any clinical/research setting where dural puncture had been used in participants of all ages and both genders, which compared divergent tip designs or diameters for prevention of PDPH.

Chapter 4

FLOW CHART

4.1Methodology



4.2 The development and use of a **special purpose gauge for crankshafts** involves several critical steps, ranging from design and material selection to manufacturing, calibration, and testing. A special purpose gauge for crankshafts must be tailored to measure critical features of the crankshaft such as **journal diameter**, **crankpin diameter**, **taper**, **roundness**, **run-out**, and **surface finish**. The methodology for creating and using such gauges focuses on ensuring **high precision**, **efficiency**, and **reliability** in the measurement process.

Key Phases in the Methodology

1. Requirement Analysis and Design Specifications

- **Objective**: To understand the specific features of the crankshaft that need to be measured and determine the tolerances required for the application.
- **Steps**:
- **Identify the Crankshaft Features to be Measured**: Critical features like journal and crankpin diameters, ovality, taper, surface roughness, and alignment need to be defined.
- **Understand Tolerances**: The acceptable limits of these features should be determined based on the crankshaft's application (e.g., automotive engine, industrial engine).
- **Review Design Drawings**: Analyse the crankshaft's design drawings or CAD models to identify key measurement points.
- **Gauge Design Concept**: Define whether **coordinate measuring system (CMM)** is appropriate, based on the measurement requirements.

2. Design of the Gauge

- **Objective**: To create a gauge that can efficiently measure the required features of the crankshaft with high accuracy and repeatability.
- **Steps**:
- **Type of Gauge Selection**:
    1. For measuring run-out or alignment: **Dial gauges** or **rotary tables**.
    2. For measuring surface roughness: **Surface finish gauges**.
- **Material Selection**: The materials used for the gauge should be durable and stable over time, commonly using **hardened steel**, **carbide**, or **ceramics** for wear resistance.
- **Tolerance Consideration**: The gauge should be able to measure within a fraction of a micron to detect any subtle deviations in the crankshaft geometry.

- **Automation and Features**: Consider integration with automation systems (like CNC or robotic arms) for faster and more consistent measurements. The design should also consider ease of use and quick feedback (e.g., **Go/No-Go** indicators).
- **Ergonomics**: The gauge should be designed to ensure ease of use, particularly in high-volume manufacturing environments, with user-friendly mechanisms for calibration and inspection.

3. Manufacturing and Fabrication of the Gauge

- **Objective**: To fabricate the gauge with precise manufacturing techniques to ensure its accuracy and long-term reliability.
- **Steps**:
- **Precision Machining**: Use **CNC machines**, **EDM (Electrical Discharge Machining)**, or **grinding** techniques to achieve the required precision in dimensions and geometric form.
- **Assembly**: If the gauge involves multiple parts (e.g., dial gauge systems or Go/No-Go mechanisms), assemble all components to form a complete measurement system.
- **Surface Finishing**: Ensure the gauge surface is smooth and free of defects that could affect measurement accuracy. This is particularly important for components that will make contact with the crankshaft, such as the journal and pin measuring surfaces.

4. Calibration of the Gauge

- **Objective**: To ensure that the special purpose gauge provides accurate and reliable measurements.
- **Steps**:
- **Use of Master Standards**: Calibrate the gauge against known standards, such as certified reference gauges, **gage blocks**, or **calibration rings**, which have been accurately measured and verified.
- **Adjusting the Gauge**: If necessary, adjust the gauge for **zeroing** or fine-tuning to account for manufacturing tolerances.
- **Verification of Tolerances**: Check the gauge's accuracy by measuring parts with known dimensions to verify that it provides consistent and correct readings.

5. Implementation in the Production Line

- **Objective**: To integrate the special purpose gauge into the manufacturing and quality control process for regular use.
- **Steps**:
- **Training Operators**: Train operators to use the special purpose gauge correctly and interpret the readings properly. This includes understanding when a part passes the measurement and when it must be rejected.
- **Incorporation into the Inspection Workflow**: Introduce the gauge into the production line where crankshafts are being manufactured. This could involve **in-process inspection** where the gauge checks parts at multiple stages (e.g., after machining of journals or crankpins).
- **Speed and Efficiency**: Ensure the measurement process is quick, especially in high-volume manufacturing environments. The use of **Go/No-Go** gauges allows for rapid assessment, while dial gauges or more complex systems might require more time for detailed measurements.

6. Data Collection and Analysis

- **Objective**: To analyse the data gathered from the special purpose gauges to identify trends, defects, and areas for improvement.
- **Steps**:
- **Recording Measurement Data**: Implement data logging and tracking systems to collect measurement results from the gauge. Modern gauges may come with digital interfaces to record data directly into a computer or database.
- **Analysis of Results**: Analyse data trends, including deviations from nominal dimensions, to identify patterns of error or variations caused by machine tool wear, environmental factors, or other issues.
- **Feedback and Continuous Improvement**: Use this data to drive continuous improvement in the manufacturing process. If consistent defects are found, they can be addressed in the machine setup, tooling, or process parameters.
- **Regular Calibration**: Periodically calibrate the gauge against certified reference standards to ensure it maintains measurement accuracy over time.

Chapter 5

5.1 3D MODEL

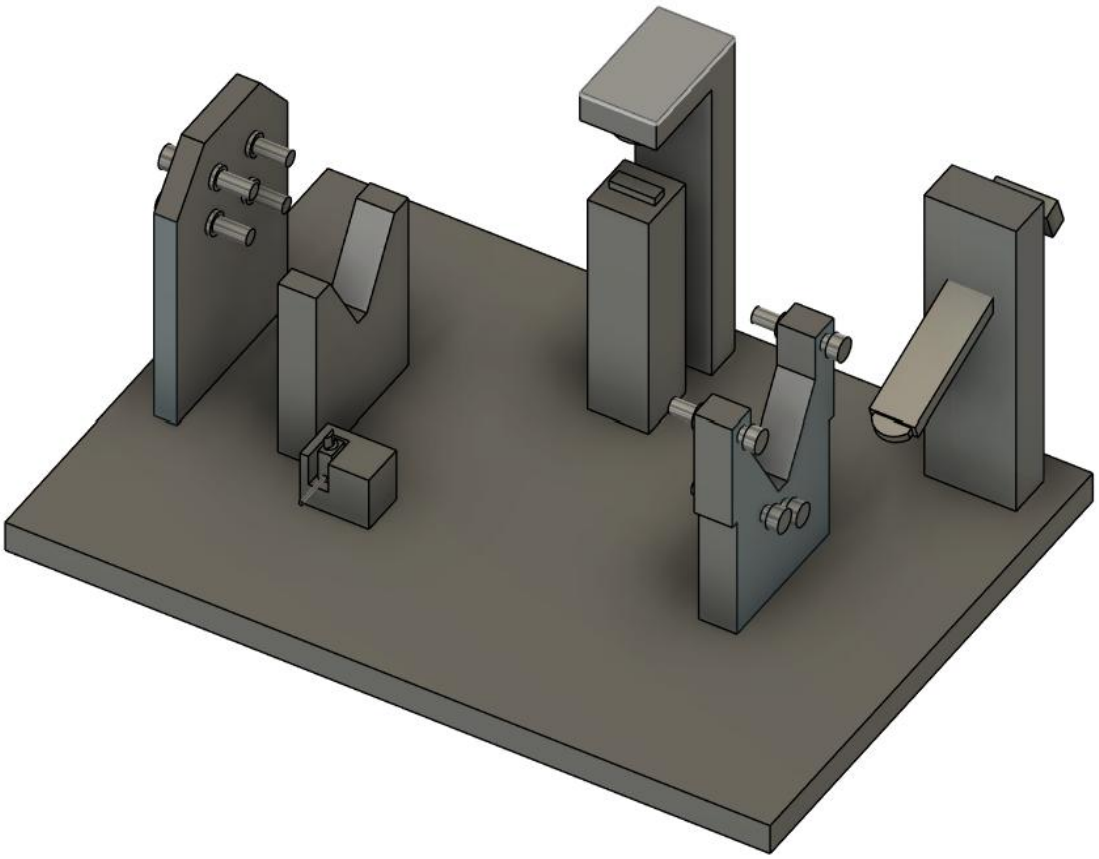


Figure 5.1:Fixture Gauge

5.2 2D DRAWING

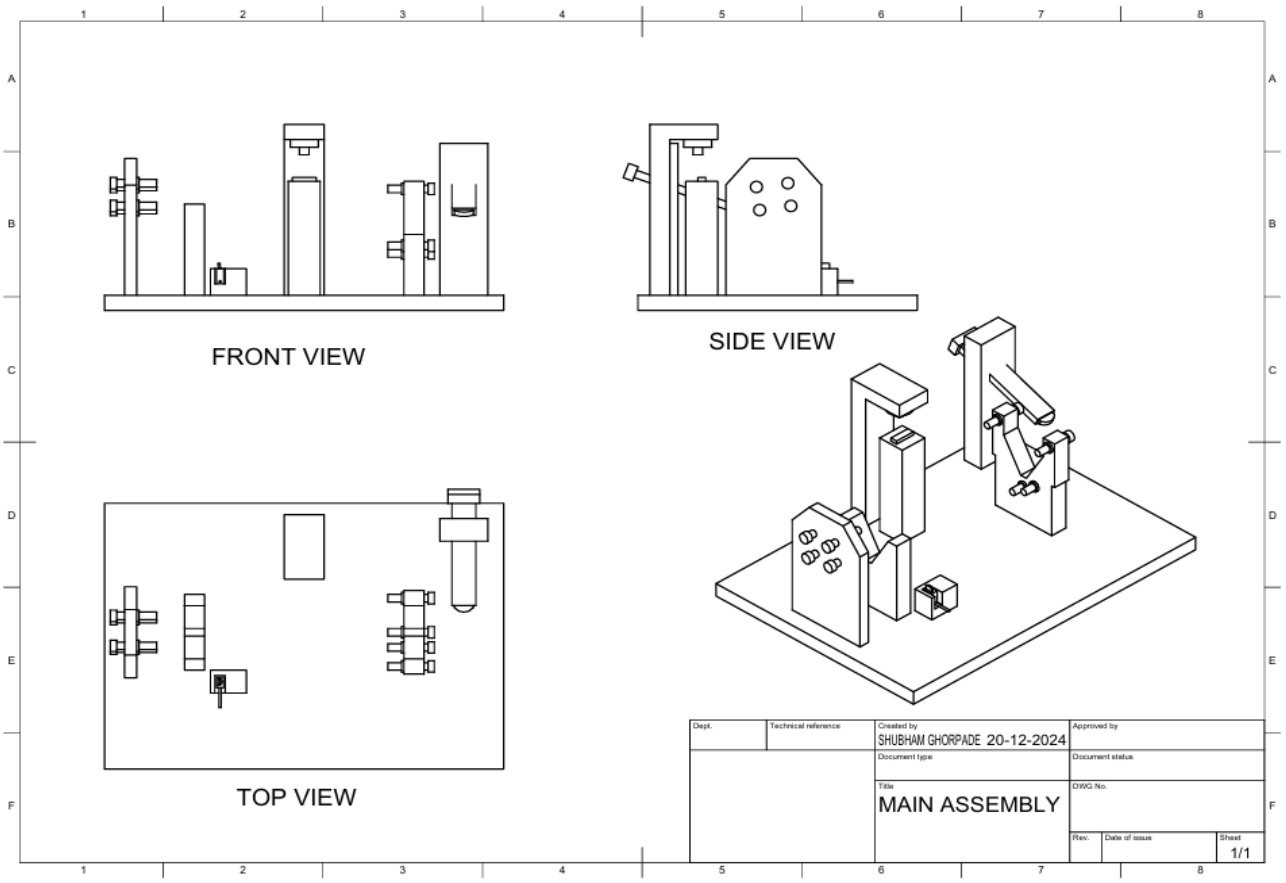


Figure 5.2:Fixture Gauge

5.3 PART 1

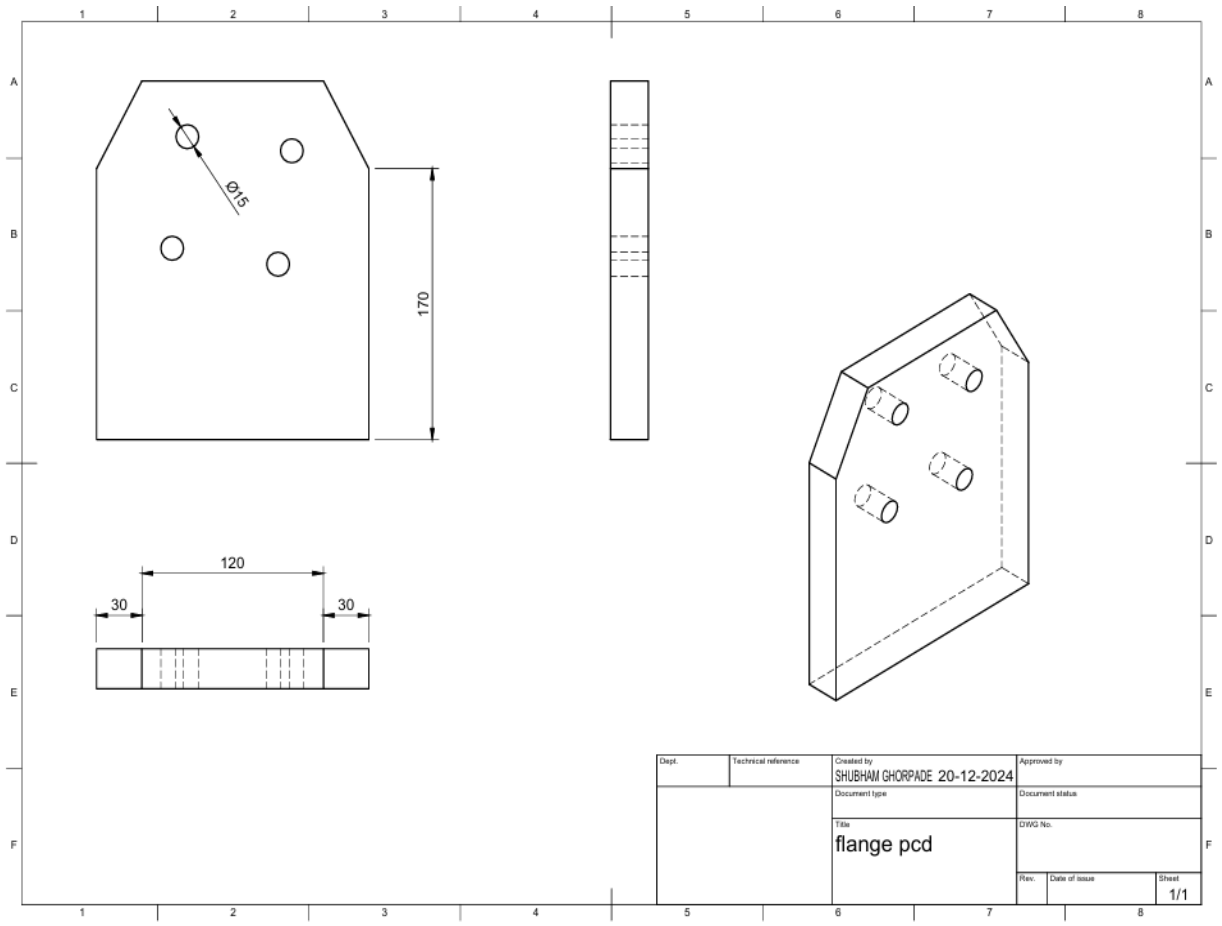


Figure 5.3:Flange PCD

5.4 PART 2

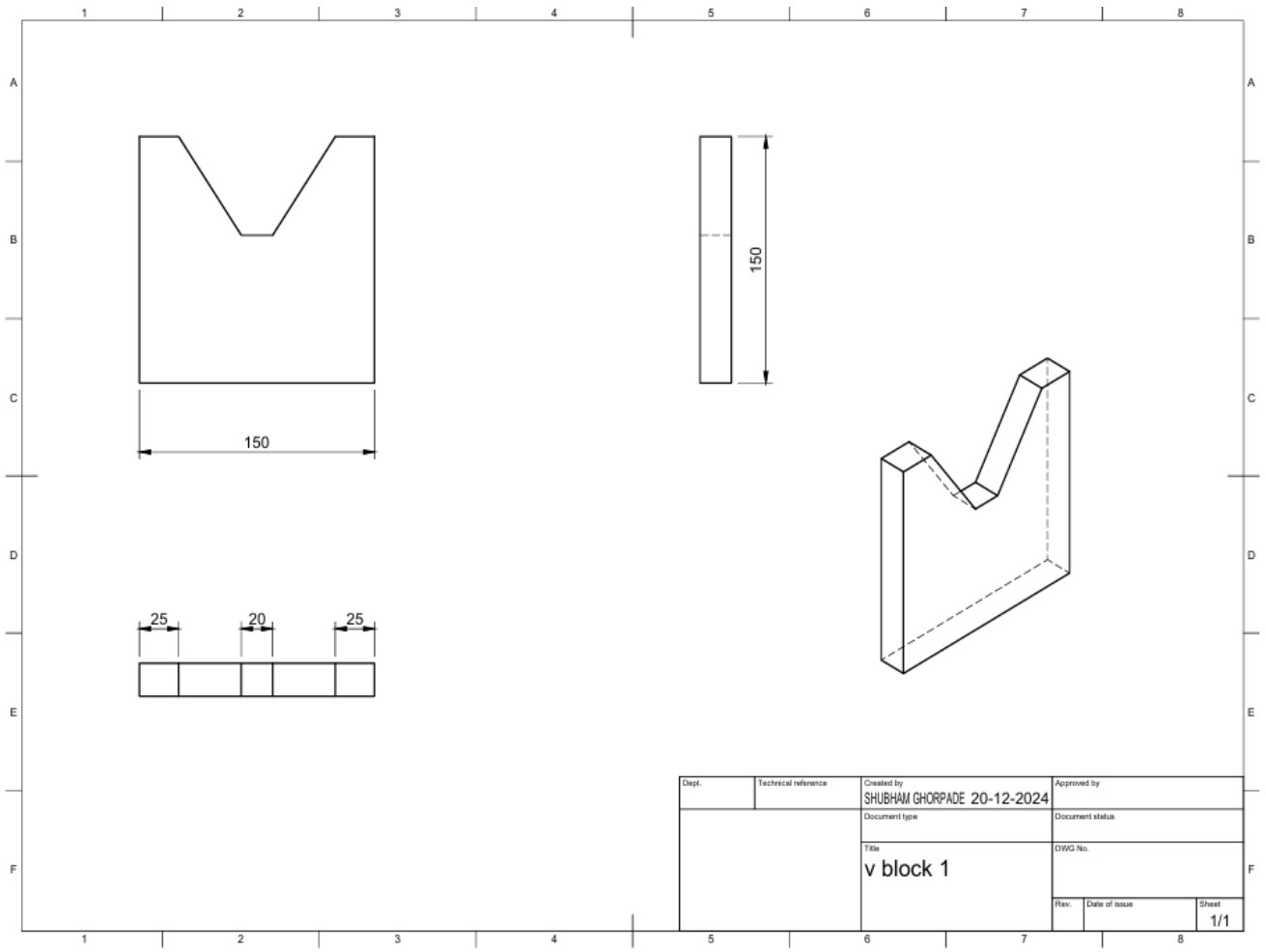


Figure 5.4:V BLOCK

5.5 PART 3

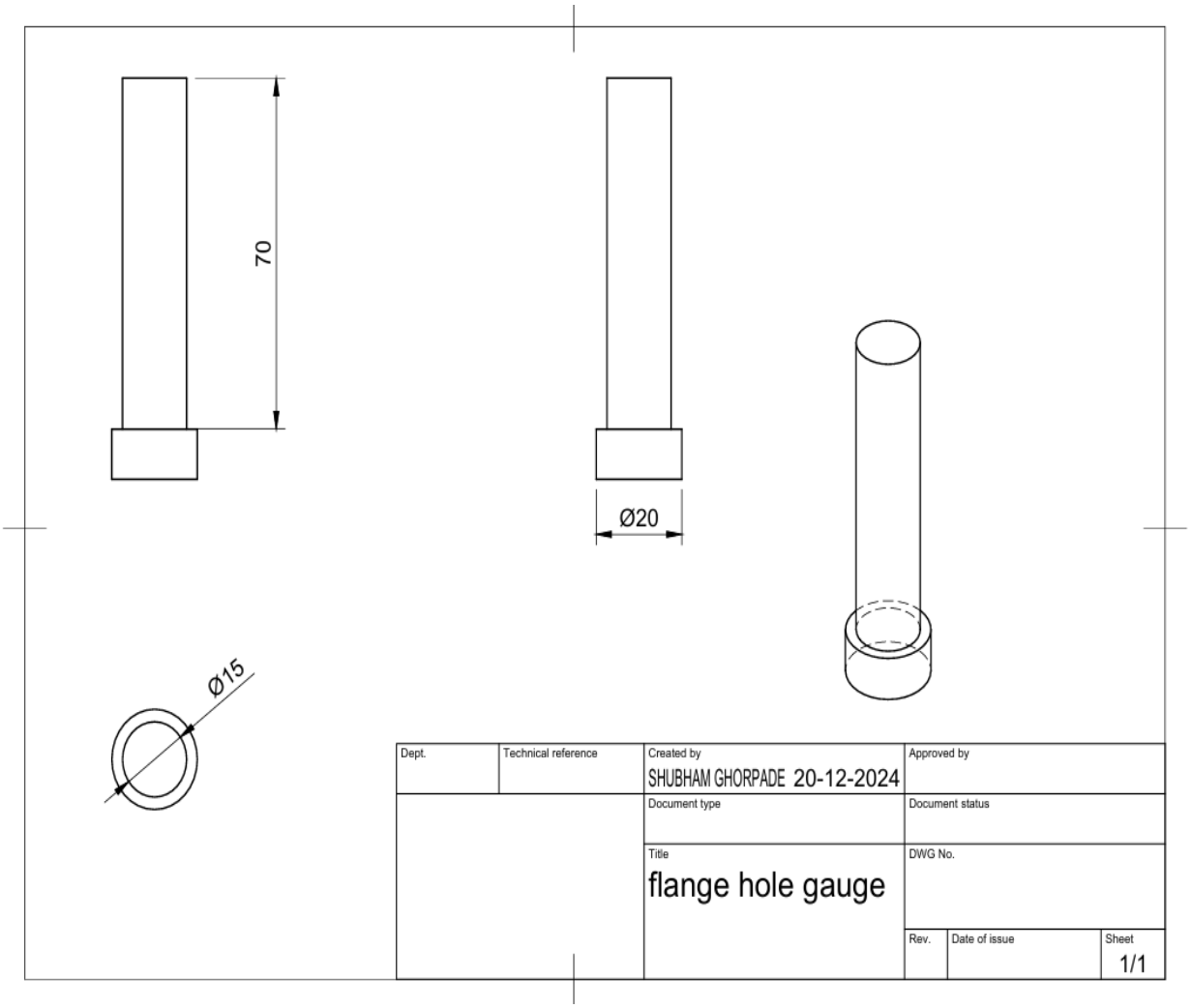


figure 5.4:Diameter gauge

5.6 PART 4

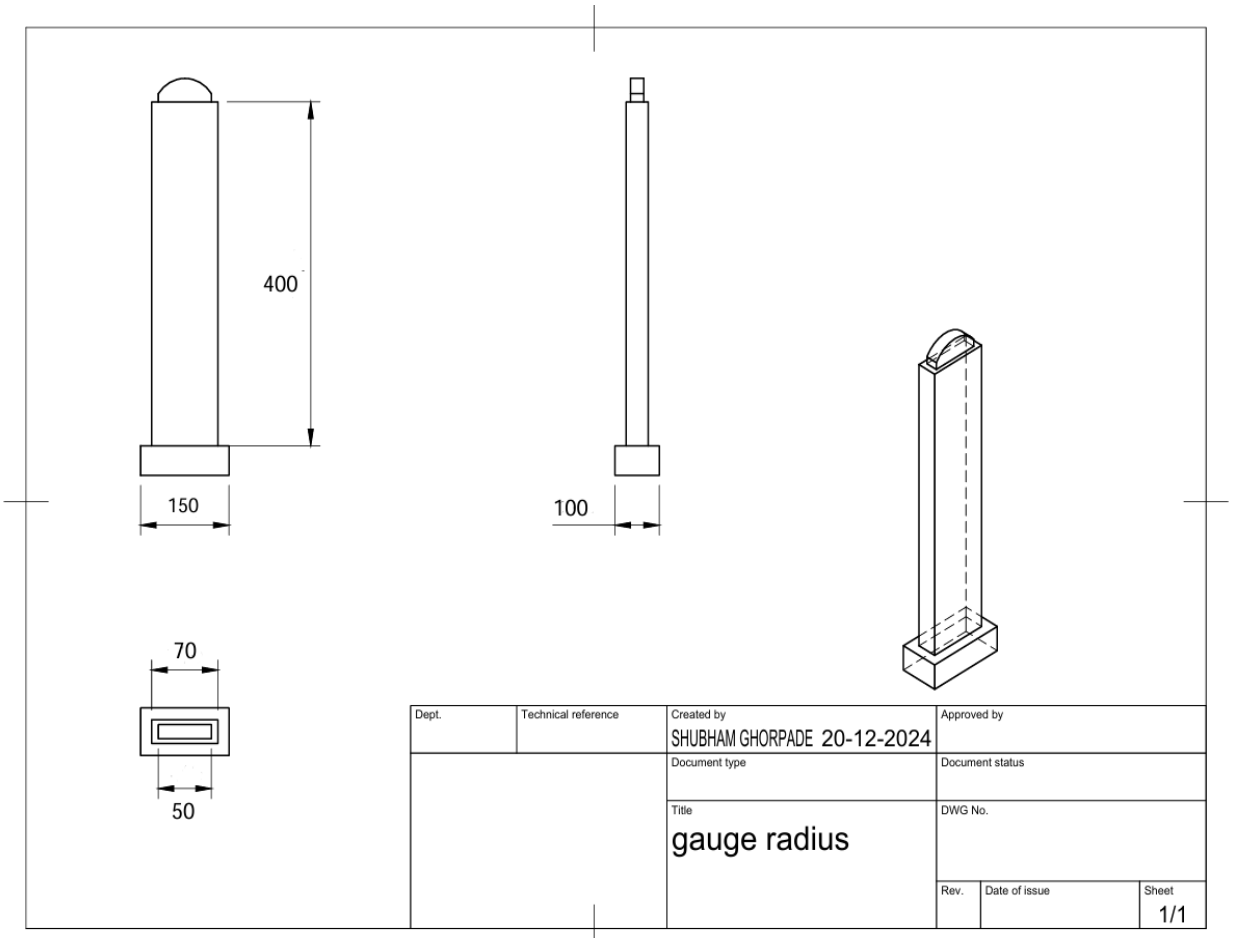


figure 5.5:Keyway Radius Gauge

Chapter 6

CONCLUSION

6.1 Conclusion

A special-purpose gauge for crankshafts plays a crucial role in ensuring the precision, quality, and performance of engine components. By providing accurate measurements of critical surfaces, such as journals and bearing points, it helps to detect any deviations or wear that could affect the functionality of the crankshaft in an engine. Here are the key conclusions regarding the use of special-purpose gauges for crankshafts.

In conclusion, a crankshaft gauge is an essential tool for accurately measuring the dimensions and tolerances of crankshafts, ensuring they meet the required specifications for performance and durability. It plays a crucial role in identifying any wear, damage, or out-of-spec conditions that could lead to engine failure or reduced efficiency. By providing precise measurements of parameters such as diameter, roundness, and taper, crankshaft gauges enable machinists and technicians to make necessary adjustments or replacements, thereby extending the life of the engine and improving overall reliability. Accurate measurements with a crankshaft gauge are vital for maintaining the integrity of the engine's operation and preventing costly repairs or premature engine failure.

6.2 future scope

In summary, special-purpose gauges for crankshafts are indispensable tools in modern manufacturing. They enhance precision, streamline production, and contribute to the overall reliability and performance of engines. By ensuring the correct dimensions and quality, these gauges help to meet the increasing demands for performance and durability in automotive and industrial machinery.

The future scope of crankshaft gauges is vast, driven by advancements in automotive technology and manufacturing processes. With the increasing demand for fuel-efficient and high-performance engines, precision in crankshaft measurement is becoming more critical. Future developments may focus on integrating smart sensors and IoT capabilities into crankshaft gauges, enabling real-time data collection and monitoring during production. Automation and AI-driven analytics are likely to enhance the accuracy and speed of measurements, reducing human error and production downtime. Moreover, the adoption of lightweight materials and complex designs in modern engines will necessitate the evolution of gauges to accommodate varying specifications and tighter tolerances.

REFRENCES

1. Ozlem Sanvir and Sanaye Umit Oktay Fitrat Marmara University, Industrial Engineering Department, Faculty of Engineering, Gazette Campus, Kadikoy-34720 Istanbul, Turkey Received: 5 August 2010 / Accepted: 20 August 2010.

2. Joel Arumun1*, Victor Ekanem1, Evans Ashigwuike2 1Alumni, Control Systems and Instrumentation, University of Huddersfield, United Kingdom 2Department of Electrical and Electronics Engineering, University of Abuja, Nigeria DOI: | Received: 13.06.2024 | Accepted: 18.07.2024 | Published: 29.08.2024.

3. Ochoa-Rodriguez, S., Wang, L.-P., Willems, P., & Olof, C. (2019). A review of radar-rain gauge data merging methods and their potential for urban hydrological applications. Water Resources Research, 55, 6356–6391Received 17 MAY 2018 Accepted 18 JUN 2019 Accepted article online 27 JUN 2019 Published online 2 AUG 2019.

4. a Department of Industrial Engineering and Systems Management, Feng Chia University, 100, Wenham Road, Seetzen, Taichung 40724, Taiwan b Department of Industrial Engineering and Management, National Chiao Tung University, Taiwan c School of Science and Engineering, George Washington University, Washington, DC, USA Received 11 April 2007 Accepted 2 January 2009 Available online 10 December 2008.

5. Evangelos Cholangitis, MD, Marco Senzel, MD, Richard Standish, MD, Laura Marelli, MD, Alberto Quaglia, MD, David Patch, MD, Amar P. Dhillon, MD, Andrew Kenneth Burroughs, MD *American Journal of Clinical Pathology*, Volume 125, Issue 5, May 2006.

6. : Arevalo-Rodriguez I, Muñoz L, Godoy-Casabianca's N, Ciapponi A, Arevalo JJ, Bogart S, Roque I Fugals M. Needle gauge and tip designs for preventing post-dural puncture headache (PDPH). Cochrane Database of Systematic Reviews 2017, published in Issue 12, 2017.

7. a School of Industrial Engineering, College of Engineering, University of Tehran, Tehran, Iran; b Arts et Métiers Paris Tech, LCFC, 57078 Metz, France; c IMT Atlantique, Lab-STICC, UBL, F-29238 Brest, France; d Universal Scientific Education and Research.

8. 15. J. Antony, M. Kumar, M.K. Tiwari, an application of six sigma methodology to reduce the engine-overheating problem in an automotive company, J. Eng. Manuf. B8 14, 633-646 (2005).